



Investments

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Group 14:

Alexandre VALLET

Leonard RAMBEAU

Victor LEGRAND

Loric REY

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1 Introduction

This report studies currency excess returns and systematic trading strategies across the G10 foreign exchange (FX) market.

The report has two main objectives. First, it characterises the statistical properties of individual currency excess returns. Second, it evaluates three standard currency factor strategies, carry, momentum, and dollar, and asks whether their premia are statistically reliable, economically meaningful, and distinct from one another.

We then move from individual factors to portfolio construction: in-sample mean-variance portfolios in Section 7, and out-of-sample conditional mean-variance efficient (CMVE) portfolios in Section 8.

Why FX? Currency markets offer a distinct testing ground for asset pricing. Unlike equities, FX excess returns are zero-sum across investors and tightly linked to interest rate differentials and exchange rate dynamics. The persistence of carry premia, in particular, constitutes a well-documented violation of Uncovered Interest Rate Parity (UIP) and lies at the heart of the “forward premium puzzle”.

Sample. We study six G10 currencies against the US dollar (AUD, CAD, EUR, GBP, JPY, NZD) over the period January 1995 to December 2024, with December 1990 as the earliest data point for signal construction and a holdout window from January 1991 through December 1994 used to initialise out-of-sample estimators. All results are reported on a monthly basis and annualised where appropriate.

2 Data

2.1 FX Rates

Spot and one-month forward mid-quote exchange rates are sourced from WRDS Datastream (`tr_ds_equities.ds2fxrate`), which stores FX series in its equities database.

We use the `exrateintcode` identifiers specified in the project brief (e.g. 2427 for AUD spot, 2601 for AUD forward) for the six currency pairs. Data are available at daily frequency from December 1990 to December 2024. To obtain a monthly panel, we take the last daily observation within each calendar month.

All rates are expressed in USD per unit of foreign currency (USD/FCU), series originally quoted in FCU/USD (e.g. CAD, JPY) are inverted, and series with inconsistent quoting conventions across spot and forward legs are handled on a per-series basis.

2.2 Interest Rates

Monthly short-term interest rates are downloaded from FRED using the OECD MEI series `IRSTCI01`, which reports call money / overnight interbank rates, annualised, not seasonally adjusted, in percentage points per year.

The seven series used are:

- USD (`IRSTCI01USM156N`), AUD (`IRSTCI01AUM156N`), CAD (`IRSTCI01CAM156N`),
- GBP (`IRSTCI01GBM156N`), JPY (`IRSTCI01JPM156N`), NZD (`IRSTCI01NZM156N`),
- a spliced EUR series using the German DEM rate (`IRSTCI01DEM156N`) before January 1999 and the Euro Area rate (`IRSTCI01EZM156N`) from January 1999 onward.

All rates are converted from annualised percentage points to monthly decimals by dividing by 1200, as required by the excess return formula.

2.3 Final Panel

The merged panel contains 409 monthly observations from December 1990 to December 2024, with 19 columns: spot rates S_t^i , one-month forward rates F_t^i , and risk-free rates r_t^i for the six

foreign currencies, plus the USD rate r_t^{US} . The panel is complete with no missing values.

3 Currency Excess Returns

3.1 Construction

The excess return in USD from a \$1 investment in currency i during month $t + 1$ is:

$$X_{t+1}^i = \frac{S_{t+1}^i}{S_t^i} (1 + r_t^i) - (1 + r_t^{US}) \quad (1)$$

This represents the payoff from borrowing \$1 at the US rate r_t^{US} , converting into $1/S_t^i$ units of foreign currency, investing at the foreign rate r_t^i , and reconverting at the realised spot rate S_{t+1}^i one month later.

The excess return thus combines two components: the **interest rate differential** ($r_t^i - r_t^{US}$) and the **exchange rate change** ($S_{t+1}^i/S_t^i - 1$).

Excess returns are computed over the full sample January 1991 to December 2024 (408 observations). All performance statistics below are reported on the **evaluation sample**: January 1995 to December 2024 (360 observations).

3.2 Summary Statistics

Table 1 (Appendix A) reports summary statistics for the six individual currency excess returns over the evaluation sample. Currencies are ordered alphabetically.

3.3 Discussion

Several patterns emerge from Table 1.

Cross-sectional heterogeneity in mean returns. Mean excess returns differ meaningfully across currencies, ranging from -3.30% for JPY to $+2.26\%$ for NZD.

This dispersion is closely aligned with the cross-section of foreign risk-free rates: AUD ($\bar{r}^i = 3.97\%$) and NZD ($\bar{r}^i = 4.46\%$) are the highest-yielding currencies and earn the largest positive excess returns, while JPY ($\bar{r}^i = 0.14\%$) is the lowest-yielding and earns the most negative excess return.

This pattern is qualitatively consistent with a carry premium: high-rate currencies tend to outperform low-rate currencies on a risk-adjusted basis, violating the UIP prediction that exchange rate changes should exactly offset rate differentials.

Statistical insignificance at the individual currency level. Despite the cross-sectional dispersion in means, no individual currency excess return is statistically distinguishable from zero at conventional thresholds.

The highest t -statistic in absolute value belongs to JPY (-1.69), which still falls short of the 1.96 threshold for 5% significance. This reflects the high idiosyncratic volatility of single-currency positions (annual standard deviations ranging from 8.1% to 12.2%) relative to their modest average returns.

This is why diversification across currencies and systematic signals matter more than single-currency bets in this setting.

Fat tails and skewness. Excess kurtosis is positive for all currencies, ranging from 1.21 (EUR) to 3.25 (JPY), indicating heavier tails than a normal distribution.

The skewness pattern is noteworthy: JPY displays positive skewness ($+0.658$), reflecting the occasional sharp yen appreciations during global risk-off episodes (carry unwind events, e.g. 2008, 2020). High-yield currencies AUD, CAD, and GBP display mild negative skewness, consistent with the well-known crash risk faced by carry investors who are long these currencies.

4 Carry Strategies

4.1 Motivation and Signal

The carry trade exploits the empirical failure of UIP. Rather than depreciating by the interest rate differential as UIP predicts, high-rate currencies tend to appreciate or remain stable, generating excess returns for the carry investor.

The signal for carry strategies is the interest rate differential:

$$\Delta r_t^i = r_t^i - r_t^{US} \quad (2)$$

4.2 Strategy Construction

CS-CARRY (Cross-Sectional Carry). Each month, currencies are ranked by Δr_t^i (rank 1 = lowest differential, rank $N = 6$ = highest). Portfolio weights are assigned as

$$w_t^i = Z \left(\text{Rank}_t^i - \frac{N+1}{2} \right) \quad (3)$$

where Z is chosen such that $\sum_{i: w_t^i > 0} w_t^i = 1$, making the strategy self-financing with exactly \$1 long and \$1 short. The strategy bets on the relative performance of high-rate versus low-rate currencies within the cross-section.

TS-CARRY (Time-Series Carry). Each currency is traded independently based on the sign of its own differential:

$$w_t^i = \frac{\text{sign}(\Delta r_t^i)}{N}. \quad (4)$$

This strategy takes a $\$1/N$ position in each currency independently of the others, going long if the foreign rate exceeds the US rate and short otherwise.

All signals are formed at month-end t and used to trade the following month's return X_{t+1}^i , with no look-ahead bias.

4.3 Results: Carry

Performance statistics for CS-CARRY and TS-CARRY, along with all other factor strategies, are reported in Table 2 (Appendix A).

Significance. Both carry strategies deliver statistically significant positive mean returns. CS-CARRY earns 3.69% per annum ($t = 2.17$) with a Sharpe ratio of 0.40, while TS-CARRY earns 2.77% ($t = 2.57$) with a Sharpe ratio of 0.47. Both t -statistics exceed the 1.96 threshold.

TS has a higher Sharpe than CS. TS-CARRY achieves a higher Sharpe ratio despite its lower mean, because its volatility (5.91%) is much lower than CS-CARRY's (9.33%). This reflects the fact that TS-CARRY positions are sized equally across currencies regardless of the magnitude of their differential, which limits concentration risk.

4.4 CS-CARRY Portfolio Weights Over Time

Figure 1 plots the CS-CARRY portfolio weight on each currency over time. The weights reveal strong persistence:

- AUD and NZD are persistently long throughout the sample, consistent with their structurally high interest rates relative to the US.
- JPY is persistently short, reflecting its near-zero rate environment since the mid-1990s.
- CAD, EUR, and GBP occupy intermediate ranks and switch sides as their monetary policy cycles diverge from or converge with the US.

An exception occurs from 2012 to 2017, when the Federal Reserve's zero-interest-rate policy (ZIRP) compressed the differential between the US rate and JPY, temporarily reducing JPY's short position.

4.5 Forward Discount and Interest Rate Differential

The net forward discount is defined (in logs) as

$$fd_t^i = \log \frac{S_t^i}{F_t^i} \quad (5)$$

By Covered Interest Rate Parity (CIP), which holds very precisely in G10 markets via forward market arbitrage:

$$\frac{F_t^i}{S_t^i} = \frac{1 + r_t^{US}}{1 + r_t^i} \implies fd_t^i = \log(1 + r_t^i) - \log(1 + r_t^{US}) \approx \Delta r_t^i \quad (6)$$

The log-linear approximation holds for small rates. Table 3 (Appendix A) reports the time-series correlation between fd_t^i and Δr_t^i for each currency.

The cross-sectional average correlation of 0.822 is high and positive, consistent with CIP theory.

Deviations from unity arise from two sources: the log-level approximation error (negligible for typical G10 rate levels), and minor data-source timing differences between Datastream forward quotes and FRED interest rate observations.

The sign is unambiguously positive: the carry signal embedded in the forward market (fd_t^i) and the interest rate differential (Δr_t^i) are two equivalent representations of the same underlying information.

5 Momentum Strategies

5.1 Motivation and Signal

Momentum strategies exploit the empirical tendency of recent currency appreciations to persist. The signal is the 12-month spot appreciation against the dollar:

$$s_{t-12,t}^i = \frac{S_t^i}{S_{t-12}^i} - 1 \quad (7)$$

This signal is constructed on the full sample starting December 1991 (the first month for which both S_t^i and S_{t-12}^i are available given our data begins December 1990).

The usual economic rationale is underreaction, meaning that exchange rates adjust to news only gradually, so past appreciation tends to predict further appreciation.

5.2 Strategy Construction

CS-MOM (Cross-Sectional Momentum). Currencies are ranked each month by $s_{t-12,t}^i$ and assigned weights following the same rank-weighting formula as CS-CARRY (Equation 3).

TS-MOM (Time-Series Momentum). Each currency is traded independently based on the sign of its own momentum signal, using the same equal-exposure formula as TS-CARRY (Equation 4).

5.3 Results: Momentum

Table 2 shows that FX momentum is much weaker than carry over this sample.

- **CS-MOM** earns only 0.24% per annum ($t = 0.15$, Sharpe = 0.03).
- **TS-MOM** earns 1.56% per annum ($t = 1.29$, Sharpe = 0.24).

Neither strategy is statistically significant at conventional thresholds. This is consistent with the broader literature, which finds that FX momentum is less robust than

equity momentum and more concentrated in specific sub-periods of trending exchange rate regimes (Menkhoff et al., 2012).

As with carry, TS-MOM has a higher Sharpe than CS-MOM because its sign-based construction produces lower volatility (6.62% vs 8.40%).

5.4 Regression of TS-CARRY on Momentum Factors

To assess whether momentum captures part of the carry premium, we estimate

$$R_{t+1}^{TS-CARRY} = \alpha + \beta_1 R_{t+1}^{CS-MOM} + \beta_2 R_{t+1}^{TS-MOM} + u_{t+1} \quad (8)$$

using OLS standard errors. Results appear in column (1) of Table 4 (Appendix A).

Carry and momentum are nearly orthogonal. In the MOM-only model, both momentum coefficients are statistically indistinguishable from zero ($t < 0.50$) and $R^2 = 0.08\%$, confirming that carry and momentum capture different information in expected return space.

The carry alpha is essentially unchanged after MOM controls. The alpha t -statistic moves only marginally, from 2.57 (standalone TS-CARRY mean) to 2.55 (after MOM controls), and the annualised alpha barely shifts (2.77% to 2.77%). The momentum betas are positive but near zero, so they absorb almost no variance from TS-CARRY. In short: adding momentum to a carry portfolio neither subsumes nor amplifies the carry premium.

6 Dollar Strategies

6.1 Motivation and Signal

Cross-sectional carry and momentum strategies are dollar-neutral by construction: they are long some currencies and short others, so any broad USD movement largely cancels out. None of the previous strategies, however, is designed to capture aggregate USD risk.

Dollar strategies instead take a common directional position on all foreign currencies simultaneously, isolating this aggregate exposure.

6.2 Strategy Construction

DOLLAR. Go long an equal-weighted basket of all N foreign currencies every month, unconditionally:

$$R_{t+1}^{DOLLAR} = \frac{1}{N} \sum_{i=1}^N X_{t+1}^i \quad (9)$$

This is a permanent bet on USD depreciation. There is no signal: the investor simply holds the foreign currency basket.

DOLLAR-CARRY. Go long (short) the same equal-weighted basket when the average interest rate differential across currencies is positive (negative):

$$R_{t+1}^{DOLLAR-CARRY} = \frac{\text{sign}(\bar{\Delta}r_t)}{N} \sum_{i=1}^N X_{t+1}^i \quad \bar{\Delta}r_t = \frac{1}{N} \sum_{i=1}^N \Delta r_t^i \quad (10)$$

The signal $\bar{\Delta}r_t$ is formed at month-end t and applied to the following month's returns, ensuring no look-ahead bias.

6.3 Results: All Factor Strategies

Table 2 shows the complete six-strategy summary.

DOLLAR is essentially zero. DOLLAR earns a mean excess return of virtually zero (-0.015% annualised, $t = -0.01$): over the full evaluation period, the USD neither appreciated nor depreciated significantly against the equal-weighted G10 basket.

With an annualised volatility of 7.58% and a Sharpe ratio of -0.002 , the unconditional long-basket strategy offered no compensation for its substantial directional risk.

DOLLAR-CARRY improves but remains insignificant. DOLLAR-CARRY earns 2.13% annualised (Sharpe = 0.28), a meaningful improvement over the unconditional strategy. The signal switches sides almost equally over the sample (179 months long, 181 months short), confirming active conditioning.

However, its t -statistic of 1.54 falls below the 1.96 threshold: the average cross-G10 rate differential is an imprecise predictor of broad USD direction at the monthly frequency.

6.4 Extended Regression: Adding DOLLAR to Carry

The extended regression in column (2) of Table 4 adds DOLLAR as a third regressor:

$$R_{t+1}^{TS-CARRY} = \alpha + \beta_1 R_{t+1}^{CS-MOM} + \beta_2 R_{t+1}^{TS-MOM} + \beta_3 R_{t+1}^{DOLLAR} + u_{t+1} \quad (11)$$

R^2 jumps from 0.08% to 16.1% . The DOLLAR factor explains a sizeable share of TS-CARRY variance: $\hat{\beta}_3 = 0.32$ with $t = 8.24$, the only significant slope coefficient in either regression.

This positive loading reflects a structural overlap. A sign-weighted carry strategy is, by construction, persistently long on high-rate currencies (AUD, NZD, GBP) and short on the lowest-rate currency (JPY). When the USD weakens globally, all foreign currencies tend to appreciate simultaneously, mechanically lifting TS-CARRY returns.

The carry premium survives the dollar control. Despite the large R^2 increase, the carry alpha barely changes (2.77% to 2.65% annualised), and its t -statistic rises from 2.57 to **2.66** .

Controlling for dollar exposure removes variance from the carry return without reducing its expected value, making the carry premium easier to detect statistically. The evidence therefore points to a carry premium that is separate from simple USD directional exposure.

7 In-sample Portfolio Analysis

We now move from individual factors to portfolio construction. We have two sets of building blocks:

- the six basis currency excess returns $\{X_{t+1}^i\}_{i=1}^6$;
- the six factor strategies (CS-CARRY, TS-CARRY, CS-MOM, TS-MOM, DOLLAR, DOLLAR-CARRY).

For each set, we form three portfolios using different weighting schemes, then scale every portfolio to a common annualised volatility of 10% for direct Sharpe comparison.

7.1 Weighting Schemes

Equal-weight (EW). $w_i = 1/N$ for all i . Uses no signal at all and serves as a naive benchmark.

Risk-parity (RP). $w_i \propto 1/\hat{\sigma}_i$, where $\hat{\sigma}_i$ is the full in-sample standard deviation of asset i . Each asset contributes equally to total portfolio risk, only the diagonal of $\hat{\Sigma}$ is exploited.

Mean-variance (MV). $w^* \propto \hat{\Sigma}^{-1}\hat{\mu}$, where $\hat{\mu}$ and $\hat{\Sigma}$ are the full in-sample sample mean and covariance matrix. This is the tangency portfolio that maximises the in-sample Sharpe ratio.

Vol scaling. Each portfolio's raw return series is multiplied by a constant $k = 10\%/\hat{\sigma}_p$ so that the realised annualised volatility equals 10% . Since this is a constant rescaling, it does not affect the Sharpe ratio. It only allows mean returns to be compared on a common risk basis.

7.2 Results: In-sample Portfolios

Performance statistics for all six in-sample portfolios are reported in Table 5 (Appendix A).

Cumulative wealth paths for all six portfolios are displayed in Figure 2 (Appendix B).

7.3 Discussion

(c) RP-CCY vs MV-CCY: the value of full-sample moments

MV-CCY (Sharpe 0.465) performs far better than RP-CCY (Sharpe -0.010).

RP-CCY uses only the diagonal of the covariance matrix (individual volatilities). MV-CCY uses the full-sample mean vector $\hat{\mu}$ and the full covariance matrix $\hat{\Sigma}$, including all pairwise correlations. The huge gap in Sharpe ratios reflects both dimensions of the Markowitz problem: expected-return estimates and the off-diagonal covariance structure.

In particular, MV-CCY tilts toward combinations of currencies with attractive estimated mean returns once their joint covariance is taken into account. RP-CCY ignores this interaction entirely because it never uses correlations, but it also never uses the cross-currency mean estimates that enter the MV-CCY first-order condition.

(d) MV-CCY vs MV-FAC

Within each period, factor returns are linear combinations of currency returns, $F_{t+1} = A_t X_{t+1}$, but A_t is **time-varying for five of the six factors**. The weight matrices for CS-CARRY, TS-CARRY, CS-MOM, TS-MOM, and DOLLAR-CARRY are signal-dependent and change every month, only DOLLAR has constant weights $1/N$. The naive identity

$$w_F^\top F = w_F^\top A X = (A^\top w_F)^\top X$$

requires A to be constant. Once A_t depends on the information set at t , a fixed factor portfolio w_F implements a time-varying currency-side exposure $A_t^\top w_F$ that lies outside the static linear span of unconditional currency portfolios, and the inequality $\text{SR}_{\text{FAC}}^* \leq \text{SR}_{\text{CCY}}^*$ does not follow from spanning.

In-sample, MV-FAC = 0.599 > MV-CCY = 0.465.

The reversal is consistent with two compatible mechanisms. First, the factor span includes dynamic conditional exposures unavailable to a static currency MV portfolio, so there is no theoretical reason for MV-CCY to dominate. Second, the six factors were explicitly engineered to concentrate positive expected returns, which sharpens $\hat{\mu}_F$ and reduces noise in $\hat{\Sigma}_F^{-1} \hat{\mu}_F$ relative to the noisier currency-level system. Both effects favour MV-FAC.

(e) EW-CCY vs MV-FAC

EW-CCY (Sharpe -0.002) is dominated by MV-FAC (Sharpe 0.599) by an order of magnitude.

This ordering is not coincidental: it is a mathematical necessity given the choice of factor set. A key observation is that

$$\text{EW-CCY} \equiv \text{DOLLAR},$$

since investing $1/N$ in each currency is exactly the equal-weighted basket return. DOLLAR is one of the six factors used to build MV-FAC, so EW-CCY lies inside the factor span.

MV-FAC is by construction the tangency portfolio of the factor space, i.e. the maximum in-sample Sharpe achievable with any combination of the six factors. Since EW-CCY is itself a feasible factor combination (with all weight on DOLLAR), we always have

$$\text{SR}_{\text{MV-FAC}}^{\text{IS}} \geq \text{SR}_{\text{EW-CCY}}^{\text{IS}}$$

Therefore it is not possible for EW-CCY to outperform MV-FAC in-sample as long as DOLLAR (or any strategy that spans EW-CCY) is included in the factor set. The argument relies

on DOLLAR's time-invariant $1/N$ weights, in contrast to the dynamic factors discussed in part (d).

The answer therefore depends explicitly on the factor set: if DOLLAR were removed, EWCCY would no longer lie in the factor span and could in principle outperform MV-FAC in some specific sample.

8 Out-of-sample Portfolio Analysis

The in-sample portfolios of Section 7 use the full 360-month sample to estimate moments. This look-ahead inflates their performance: a real-time investor can only use data available at the formation date.

We now construct conditional mean-variance efficient (CMVE) portfolios that satisfy

$$w_t^{CMVE} = k_t \hat{\Sigma}_t^{-1} \hat{\mu}_t \quad (12)$$

where $\hat{\Sigma}_t$ and $\hat{\mu}_t$ are estimated using only data observed through month t , and k_t is a scaling factor that targets a constant ex-ante annualised volatility of 10%.

8.1 Conditional Covariance: EWMA

The conditional covariance is estimated by an exponentially weighted moving average (EWMA) with the standard RiskMetrics decay parameter $\lambda = 0.97$:

$$\hat{\Sigma}_t = (1 - \lambda) X_t X_t^\top + \lambda \hat{\Sigma}_{t-1} \quad (13)$$

This recursion gives an effective half-life of approximately 23 months. It is initialised at December 1994 with the sample covariance over the holdout window (January 1991 to December 1994, 48 observations).

8.2 Conditional Mean: Two Approaches

Expanding-window mean (Exp). A naive benchmark using all observations through t :

$$\hat{\mu}_{i,t}^{\text{Exp}} = \frac{1}{N_t} \sum_{s=1}^t X_s^i \quad (14)$$

Fama-MacBeth mean (FM). Each month s , run a cross-sectional regression of the six excess returns on the carry and momentum signals, with no intercept:

$$X_{s+1}^i = \gamma_s \Delta r_s^i + \phi_s s_{s-12,s}^i + \varepsilon_s^i \quad (15)$$

Take the expanding-window average of the slope estimates,

$$\bar{\gamma}_t = \frac{1}{K_t} \sum_{s \leq t-1} \hat{\gamma}_s \quad \bar{\phi}_t = \frac{1}{K_t} \sum_{s \leq t-1} \hat{\phi}_s$$

and apply them to the current signals:

$$\hat{\mu}_{i,t}^{\text{FM}} = \bar{\gamma}_t \Delta r_t^i + \bar{\phi}_t s_{t-12,t}^i \quad (16)$$

The FM regressions are pre-trained on the 36 cross-sections of the holdout window (December 1991 to November 1994), so the loop starts in January 1995 with non-degenerate average coefficients.

8.3 Vol Targeting

For each formation date, the scaling factor is

$$k_t = \frac{0.10}{\sqrt{\hat{\mu}_t^\top \hat{\Sigma}_t^{-1} \hat{\mu}_t \times 12}} \quad (17)$$

which makes the ex-ante annualised volatility of w_t^{CMVE} exactly equal to 10% under the conditional moments.

Realised (ex-post) volatility may differ because the conditional moment estimates are themselves noisy.

8.4 Fama-MacBeth Coefficients: Discussion (d)

We report the time-series averages of $(\hat{\gamma}_s, \hat{\phi}_s)$ over the evaluation sample only, using 360 cross-sectional estimates from December 1994 through November 2024, results are given in Table 6 (Appendix A).

How the t-stats are computed. The cross-sectional regression is re-estimated each month, yielding a time series of slopes $\{\hat{\gamma}_s, \hat{\phi}_s\}$ over $T = 360$ evaluation months.

The Fama-MacBeth t -statistic on $\bar{\gamma}$ is the time-series mean of the monthly slope estimates divided by its OLS standard error.

Statistical significance. The carry slope is significant at the 5% level (OLS $t = 2.43$), confirming a positive cross-sectional reward to high-rate currencies.

The momentum slope is not significant (OLS $t = 0.98$), echoing the weak standalone performance of CS-MOM and TS-MOM in Section 5.

Economic interpretation. A positive $\bar{\gamma}$ means that an extra unit of rate differential predicts higher next-month excess return, which is the carry premium and a violation of UIP.

A positive $\bar{\phi}$ (though small here) means that recently appreciating currencies tend to keep appreciating, which is the FX momentum premium.

Link to the factor strategies. The FM regression projects the six currency returns onto the exact same two signals used to build CS/TS-CARRY and CS/TS-MOM. The FM forecast $\hat{\mu}_t^{\text{FM}}$ is therefore a linear combination of the carry and momentum signals, with weights given by their estimated cross-sectional risk premia.

The CCV-FM portfolio thus internalises the two strongest empirical regularities of the FX cross-section in a single optimisation, rather than treating them as separate ad-hoc bets. This is the main reason CCV-FM outperforms CCV-Exp out-of-sample.

The evolution of both coefficients over the evaluation period is shown in Figure 3 (Appendix B).

8.5 CCV Portfolios: Results

Out-of-sample portfolio performance is reported in Table 7 (Appendix A).

Cumulative wealth paths for CCV-Exp and CCV-FM alongside the in-sample MV benchmarks are provided in Figure 4 (Appendix B).

8.6 Performance Evaluation: Discussion (f)

CCV-FM outperforms CCV-Exp. With a Sharpe of 0.515 versus 0.180, CCV-FM is roughly three times more efficient than CCV-Exp. The FM forecast exploits the cross-sectional carry and momentum signals, which concentrate the predictive content in a low-dimensional space.

The expanding sample mean used by CCV-Exp is very noisy, especially in the early years where only a few dozen observations are available and the FX signal-to-noise ratio is low.

CCV-FM almost matches in-sample MV-FAC. The in-sample MV portfolios use the full 360-month sample to estimate both moments, which is impossible in real time. CCV-FM, which forms portfolios using only past data, achieves a Sharpe of 0.515, within 0.08 of the in-sample MV-FAC’s 0.599.

This 0.08 gap is a reasonable estimate of the cost of operating in real time with this factor structure.

Where the OOS degradation comes from. Most of the estimation error comes from the conditional mean $\hat{\mu}_t$. Second moments converge much faster than means, so the EWMA covariance estimator is comparatively well-behaved.

CCV-FM bypasses the worst part of the mean-estimation problem by imposing a parametric structure (two factors: carry and momentum) instead of estimating six independent currency means.

Realised vs ex-ante volatility. The scaling factor targets 10% annualised vol using conditional moments. Realised volatility is 11.83% for CCV-Exp and 12.00% for CCV-FM, slightly above target due to noise in real-time parameter estimates.

Mechanically, the in-sample portfolios hit exactly 10% because they use ex-post moments.

9 Conclusion

This report studies currency excess returns and a factor-based investment pipeline in the G10 FX market from 1995 to 2024.

Individual currencies: weak signals, high noise. No single currency delivers a Sharpe ratio above 0.20 in absolute value or a significant t -statistic. This motivates systematic multi-currency strategies rather than individual bets.

Factor strategies: carry is the clearest signal. Among the six factor strategies, carry is the clearest signal. CS-CARRY ($t = 2.17$) and TS-CARRY ($t = 2.57$) both deliver statistically significant returns with Sharpe ratios approaching 0.5.

Momentum is much weaker (CS-MOM: $t = 0.15$, TS-MOM: $t = 1.29$, both insignificant) and the unconditional DOLLAR factor earns essentially zero. DOLLAR-CARRY improves on DOLLAR but does not reach significance.

The carry premium survives the controls. The carry premium survives controls for both momentum and dollar exposure: its alpha t -statistic actually rises from 2.57 standalone to 2.66 after controlling for MOM + DOLLAR. This suggests that the premium is not an artefact of momentum or USD risk.

In-sample portfolios: factors beat raw currencies. Of the six 10%-vol-scaled portfolios, MV-FAC achieves the highest Sharpe (0.599), followed by MV-CCY (0.465). Naive equal-weight currency portfolios are essentially unprofitable.

The factor space achieves better in-sample performance because the factors are designed to concentrate the cross-sectional signal.

Out-of-sample CMVE: parsimony wins. The Fama-MacBeth-based CCV-FM achieves a Sharpe of 0.515 in real time, roughly three times the naive expanding-mean CCV-Exp (0.180), and within 0.08 of the in-sample MV-FAC.

Overall, the practical value of mean-variance optimisation in FX seems to depend on imposing a parsimonious structure (two factors: carry and momentum) on the conditional mean. Estimating six independent means from monthly FX data is extremely difficult, estimating two risk premia from a cross-sectional regression is much more feasible.

Synthesis. Carry, momentum, and dollar capture distinct sources of FX risk premia, with carry being the strongest and momentum the weakest. Their combination through a Fama-MacBeth cross-sectional projection delivers a real-time Sharpe of 0.5, demonstrating that the asset-pricing intuition translates directly into implementable portfolio gains.

A Tables

Table 1: Currency Excess Returns: Summary Statistics (Jan 1995 to Dec 2024)

	AUD	CAD	EUR	GBP	JPY	NZD
Mean (% , ann.)	1.408	0.334	−0.888	0.101	−3.302	2.258
t -statistic	0.656	0.227	−0.527	0.067	−1.685	1.013
Std (% , ann.)	11.758	8.064	9.222	8.278	10.733	12.210
Sharpe ratio	0.120	0.041	−0.096	0.012	−0.308	0.185
Skewness	−0.192	−0.281	0.035	−0.206	0.658	−0.111
Excess kurtosis	1.303	2.912	1.214	1.222	3.254	1.242
N obs	360	360	360	360	360	360
$\bar{r}^i$ (% , ann.)	3.968	2.586	1.750	3.001	0.140	4.459

Mean and standard deviation are annualised (multiply monthly by 12 and $\sqrt{12}$ respectively). Sharpe ratio is annualised mean divided by annualised standard deviation. t -statistic tests $H_0 : \mathbb{E}[X^i] = 0$. Skewness and excess kurtosis are bias-corrected. $\bar{r}^i$ is the average foreign risk-free rate over the evaluation sample.

Table 2: Factor Strategies: Performance Statistics (Jan 1995 to Dec 2024)

	CS-CARRY	TS-CARRY	CS-MOM	TS-MOM	DOLLAR	DOLLAR-CARRY
Mean (% , ann.)	3.691	2.772	0.235	1.564	−0.015	2.128
t -statistic	2.166	2.568	0.153	1.293	−0.011	1.543
Std (% , ann.)	9.334	5.912	8.405	6.624	7.579	7.554
Sharpe ratio	0.395	0.469	0.028	0.236	−0.002	0.282

All statistics computed over the evaluation sample (360 monthly observations). Mean and standard deviation are annualised. t -statistic tests $H_0 : \mathbb{E}[R] = 0$.

Table 3: Time-Series Correlation between Forward Discount fd_t^i and Rate Differential Δr_t^i

	AUD	CAD	EUR	GBP	JPY	NZD	Average
$\text{Corr}(fd_t^i, \Delta r_t^i)$	0.891	0.668	0.864	0.688	0.967	0.852	0.822

Computed over the evaluation sample (January 1995 to December 2024). Cross-sectional average reported in bold.

Table 4: Regression of TS-CARRY on Momentum and Dollar Factors (OLS standard errors)

	(1) MOM only		(2) MOM + DOLLAR	
	Coef.	<i>t</i> -stat	Coef.	<i>t</i> -stat
Intercept (α)	0.0023	2.55**	0.0022	2.66***
CS-MOM (β_1)	0.0196	0.44	0.0193	0.47
TS-MOM (β_2)	0.0013	0.02	0.0802	1.51
DOLLAR (β_3)	—	—	0.3195	8.24***
R^2	0.08%		16.1%	
Standalone $t(\bar{R}^{TS-CARRY})$			2.57	

** significant at 5%; *** significant at 1%. $N = 360$ monthly observations. OLS standard errors. Intercepts are monthly, annualised alpha = $\alpha \times 12 \times 100$.

Table 5: In-sample Portfolios: Performance Statistics (Jan 1995 to Dec 2024, scaled to 10% vol)

	EW-CCY	RP-CCY	MV-CCY	EW-FAC	RP-FAC	MV-FAC
Mean (% , ann.)	−0.020	−0.102	4.652	4.081	4.154	5.989
<i>t</i> -statistic	−0.011	−0.056	2.548	2.235	2.275	3.280
Std (% , ann.)	10.000	10.000	10.000	10.000	10.000	10.000
Sharpe ratio	−0.002	−0.010	0.465	0.408	0.415	0.599

CCY = portfolios of the six basis currencies, FAC = portfolios of the six factor strategies. All portfolios scaled ex-post to 10% annualised volatility using the full in-sample standard deviation.

Table 6: Fama-MacBeth Coefficient Summary (evaluation sample, 360 cross-sections)

	Mean	<i>t</i> -stat (OLS)	Significance
$\bar{\gamma}$ (carry slope)	1.837	2.43	Yes at 5%
$\bar{\phi}$ (momentum slope)	0.014	0.98	No

OLS *t*-statistic uses the time-series variability of the monthly slope estimates ($T = 360$ evaluation months).

Table 7: Out-of-sample CMVE Portfolios vs. In-sample Benchmarks (Jan 1995 to Dec 2024)

	CCV-Exp (OOS)	CCV-FM (OOS)	MV-CCY (IS)	MV-FAC (IS)	EW-CCY (IS)
Mean (% , ann.)	2.127	6.182	4.652	5.989	−0.020
<i>t</i> -statistic	0.985	2.822	2.548	3.280	−0.011
Std (% , ann.)	11.83	12.00	10.00	10.00	10.00
Sharpe ratio	0.180	0.515	0.465	0.599	−0.002

CCV-Exp and CCV-FM are constructed in real time using only data available at each formation date. Their realised vol differs slightly from the 10% ex-ante target because of estimation noise. In-sample (IS) benchmarks reproduce columns from Table 5 for direct comparison.

B Figures

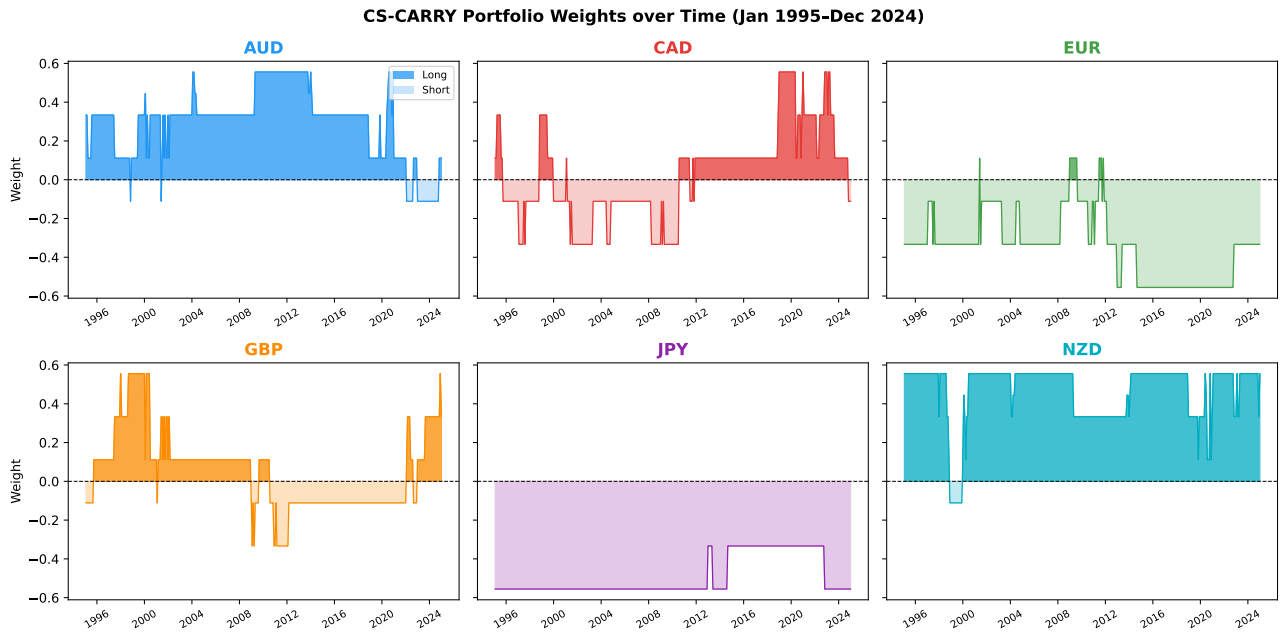


Figure 1: CS-CARRY portfolio weights on each of the six currencies (January 1995 to December 2024). Dark fill = long position, light fill = short position. AUD and NZD are persistently long (high rates), JPY is persistently short (near-zero rate).

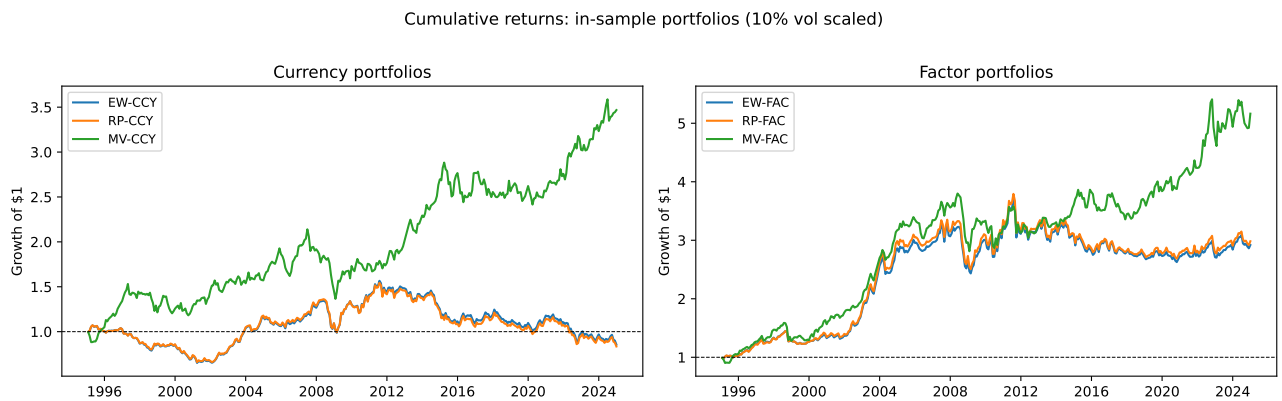


Figure 2: Cumulative returns of the six in-sample portfolios (10%-vol scaled). MV-FAC dominates throughout, EW-CCY drifts horizontally around \$1, mirroring the near-zero DOLLAR return.

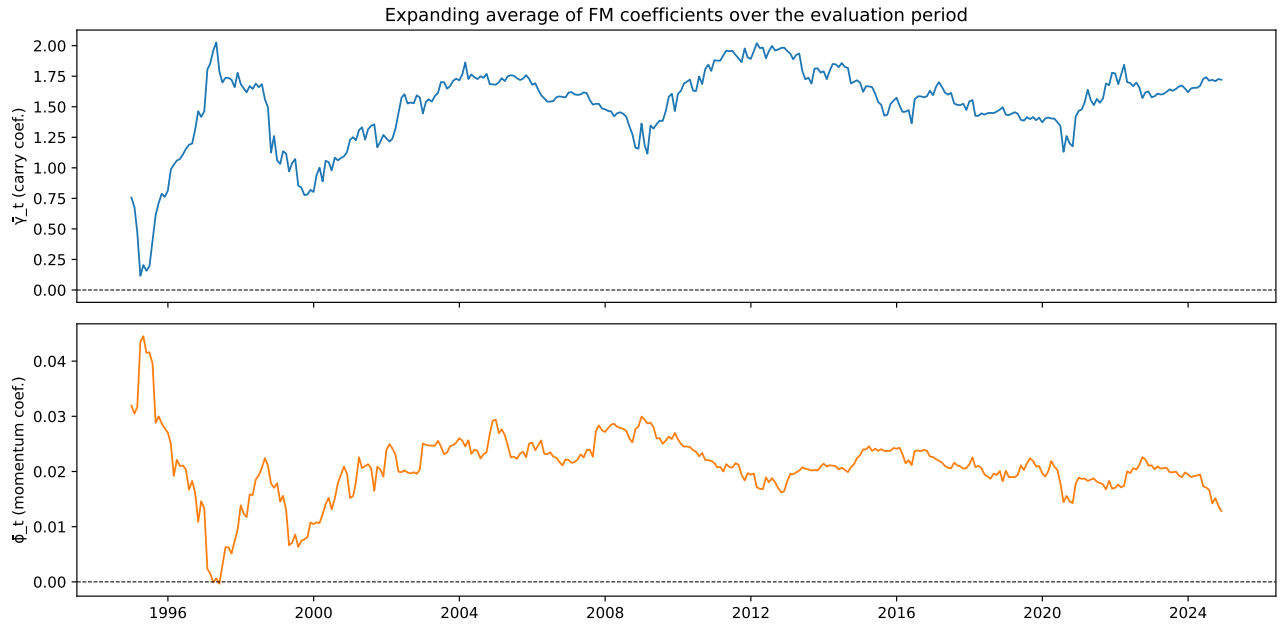


Figure 3: Evolution of the expanding-window FM coefficients $\bar{\gamma}_t$ (carry, top) and $\bar{\phi}_t$ (momentum, bottom) over the evaluation period. Both stabilise quickly after the first few years.

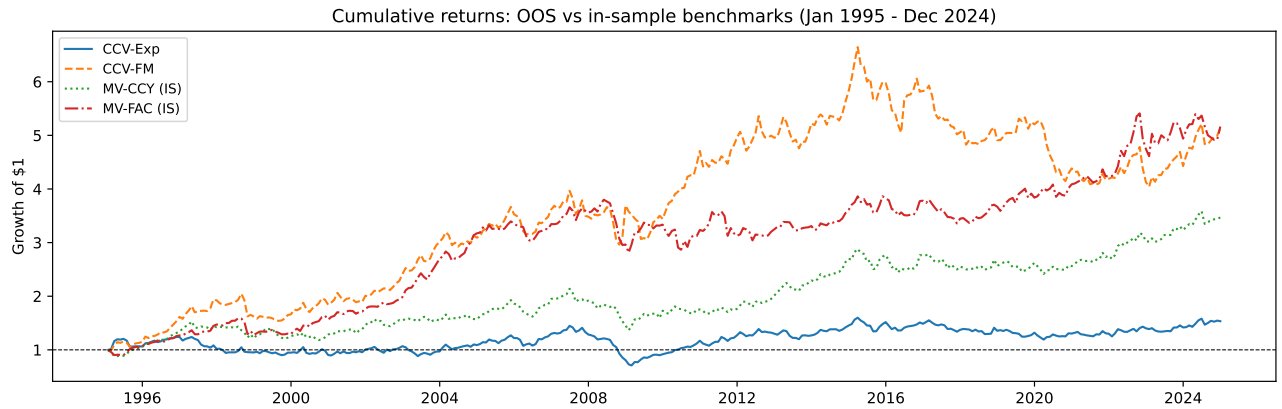


Figure 4: Cumulative returns of the OOS portfolios (CCV-Exp, CCV-FM) alongside the in-sample MV benchmarks. CCV-FM tracks the in-sample MV-FAC closely despite operating in real time.